

RSV

Training Data Format

Field	Description	Default or source
location_code	Location identifier	influpaint/influpaint_locations.csv
value	Observed/modelled values	All rates or all counts, scaled consistently
fluseason_week	Week index within season	season_week() 1 to 53
fluseason	Season start year	season_year()
datasetH1	High level dataset identifier	e.g. RSV scenario modelling hub
datasetH2	Specific dataset or scenario identifier	e.g. 2023-11-12-NIH-RSV_Phenomenological
sample	Sample identifier	e.g. 1.1, 1.2

Helpers

```
library(tidyverse)
library(arrow)

location_codes <- read_csv("influpaint/influpaint_locations.csv")

#' Get the season year for a given date
#' @param date A Date object
#' @param start_month Integer. Month defining season start. Default is June.
#' @return Integer. The season year.
#' @details
#' The start of the season is defined by `start_month` (default is June).
#' If the date is on or after `start_month`, the season year is the current year.
```

```

#' Else, it is the previous year.
season_year <- function(date, start_month = 6) {
  year <- as.integer(format(date, "%Y"))
  month <- as.integer(format(date, "%m"))
  ifelse(month >= start_month, year, year - 1)
}

#' Get the week index within the season for a given date
#' @param date A Date object
#' @param start_month Integer. Month defining season start. Default is June.
#' @return Integer. Week index within the season (1 to 53)
#' @details
#' The start of the season is defined by `start_month` (default is June).
#' The week index is calculated as the number of weeks since the season start date.
season_week <- function(date, start_month = 6) {
  sy <- season_year(date, start_month)
  season_start <- as.Date(paste(sy, start_month, 1, sep = "-"))
  as.integer(floor(as.numeric(date - season_start) / 7) + 1)
}

```

Fetch RSV Data

NSSP

National Syndromic Surveillance Program Emergency Department Visits [source](#)

pct_ed_visits_rsv : Percent of ED visits that had a discharge diagnosis code of rsv

Earliest date available: 2022-10-01

```

library(epidatr)
nssp <- pub_covidcast(
  source = "nssp",
  signals = "pct_ed_visits_rsv",
  geo_type = "state",
  time_type = "week"
) |>
mutate(geo_value = str_to_upper(geo_value)) |>
left_join(
  location_codes |>
  select(abbreviation, location_code),

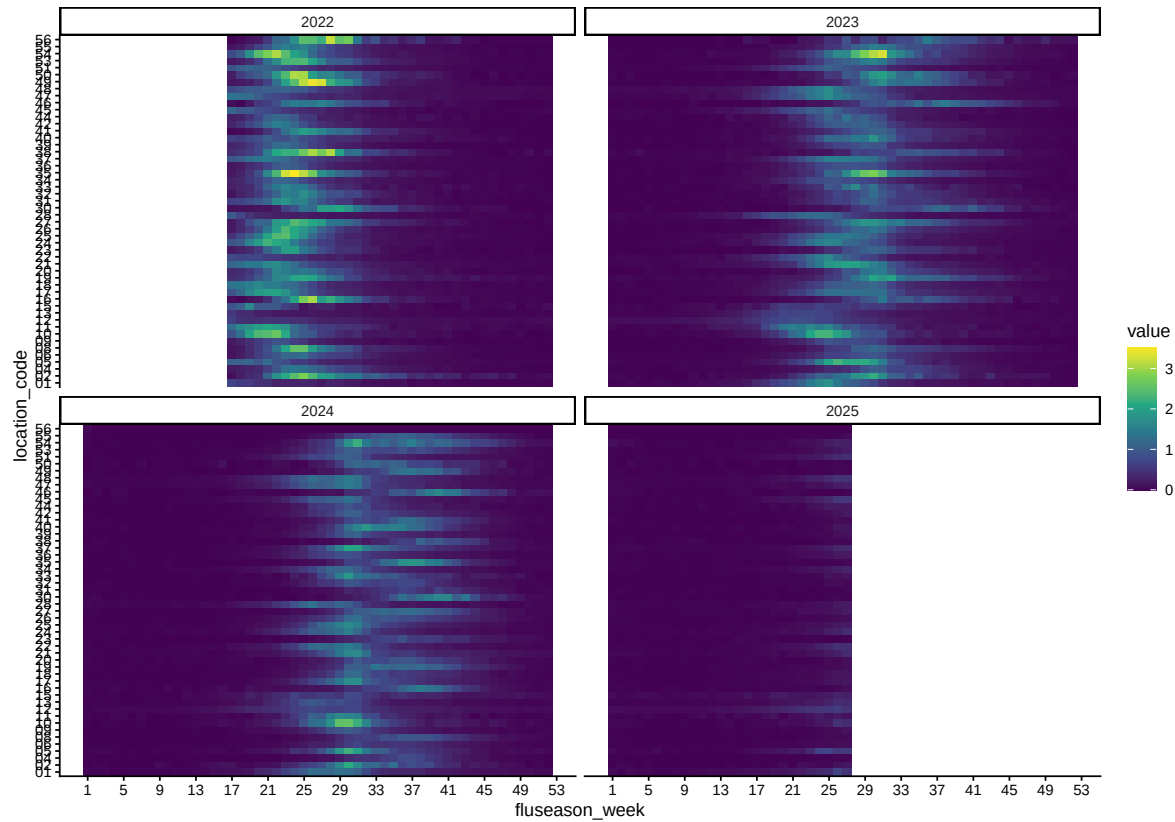
```

```

    by = c("geo_value" = "abbreviation")
  ) |>
  transmute(
    location_code,
    fluseason = season_year(as.Date(time_value)),
    fluseason_week = season_week(as.Date(time_value)),
    value,
    datasetH1 = "NSSP",
    datasetH2 = "NSSP",
    sample = "1"
  )

nssp |>
  ggplot(aes(x = fluseason_week, y = location_code)) +
  facet_wrap(~fluseason) +
  geom_tile(aes(fill = value)) +
  scale_fill_viridis_c() +
  scale_x_continuous(breaks = seq(1, 53, by = 4)) +
  theme_classic()

```



```
nssp |>
  skimr::skim() |>
  as_tibble() |>
  select(skim_variable, complete_rate)
```

```
# A tibble: 7 x 2
  skim_variable complete_rate
  <chr>          <dbl>
1 location_code      1
2 datasetH1         1
3 datasetH2         1
4 sample            1
5 fluseason         1
6 fluseason_week    1
7 value             1
```

```
write_parquet(nssp, "RSV_files/NSSP.parquet")
```

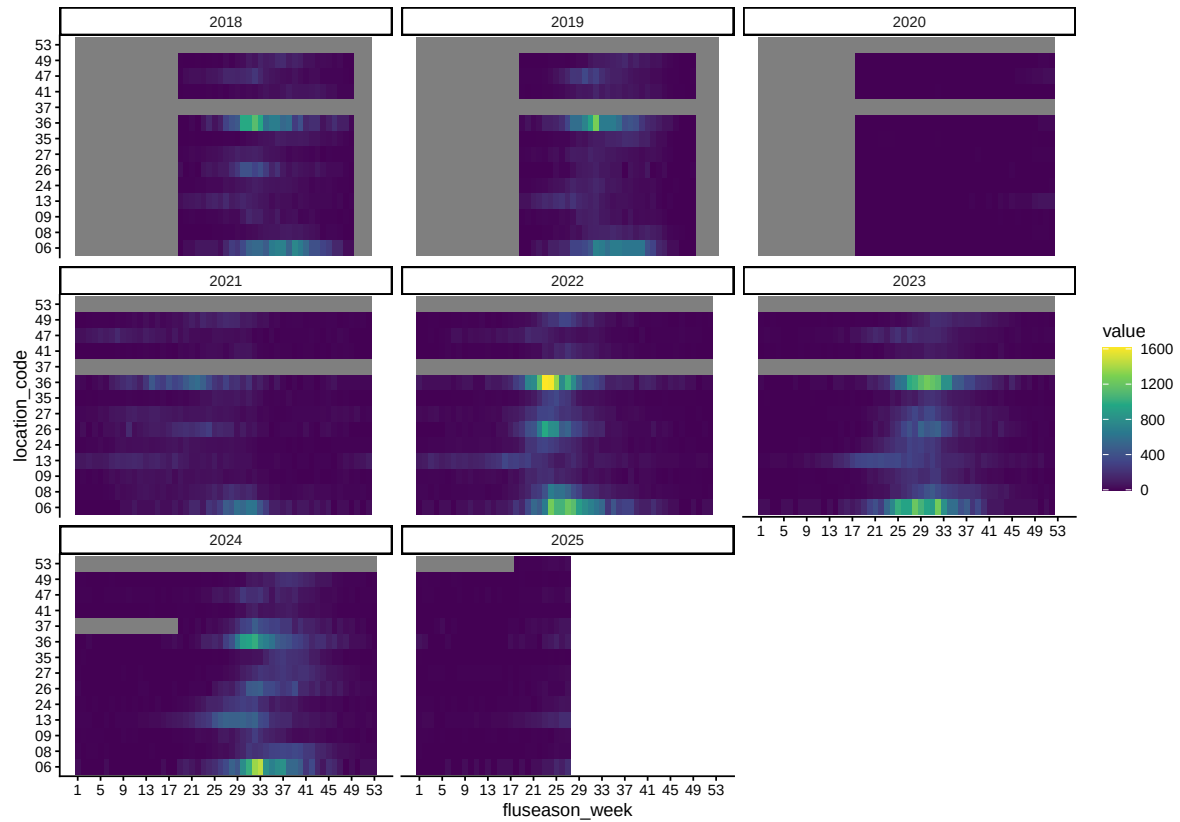
RSV-Net

Weekly Rates of Laboratory-Confirmed RSV Hospitalizations from the RSV-NET Surveillance System [source](#):

RSV scenario modeling hub [time-series.csv](#) converts to **weekly number of hospitalizations in each given state**

```
rsvnet <- read_csv(
  "https://raw.githubusercontent.com/midas-network/rsv-scenario-modeling-hub/main/target-data.csv"
) |>
  filter(
    age_group == "0-130",
    location != "US"
  ) |>
  transmute(
    location_code = location,
    fluseason = season_year(date),
    fluseason_week = season_week(date),
    value = observation,
    datasetH1 = "RSV-Net",
    datasetH2 = "RSV-Net",
    sample = "1"
  ) |>
  filter(fluseason >= 2018)

rsvnet |>
  ggplot(aes(x = fluseason_week, y = location_code)) +
  facet_wrap(~fluseason) +
  geom_tile(aes(fill = value)) +
  scale_fill_viridis_c() +
  scale_x_continuous(breaks = seq(1, 53, by = 4)) +
  theme_classic()
```



```
rsvnet |>
  skimr::skim() |>
  as_tibble() |>
  select(skim_variable, complete_rate)
```

```
# A tibble: 7 x 2
  skim_variable complete_rate
  <chr>          <dbl>
1 location_code      1
2 datasetH1         1
3 datasetH2         1
4 sample            1
5 fluseason         1
6 fluseason_week    1
7 value             0.739
```

```
write_parquet(rsvnet, "RSV_files/RSV_NET.parquet")
```

Missing values & exclude 2020?

NHSN

National Healthcare Safety Network (NHSN) [source](#)

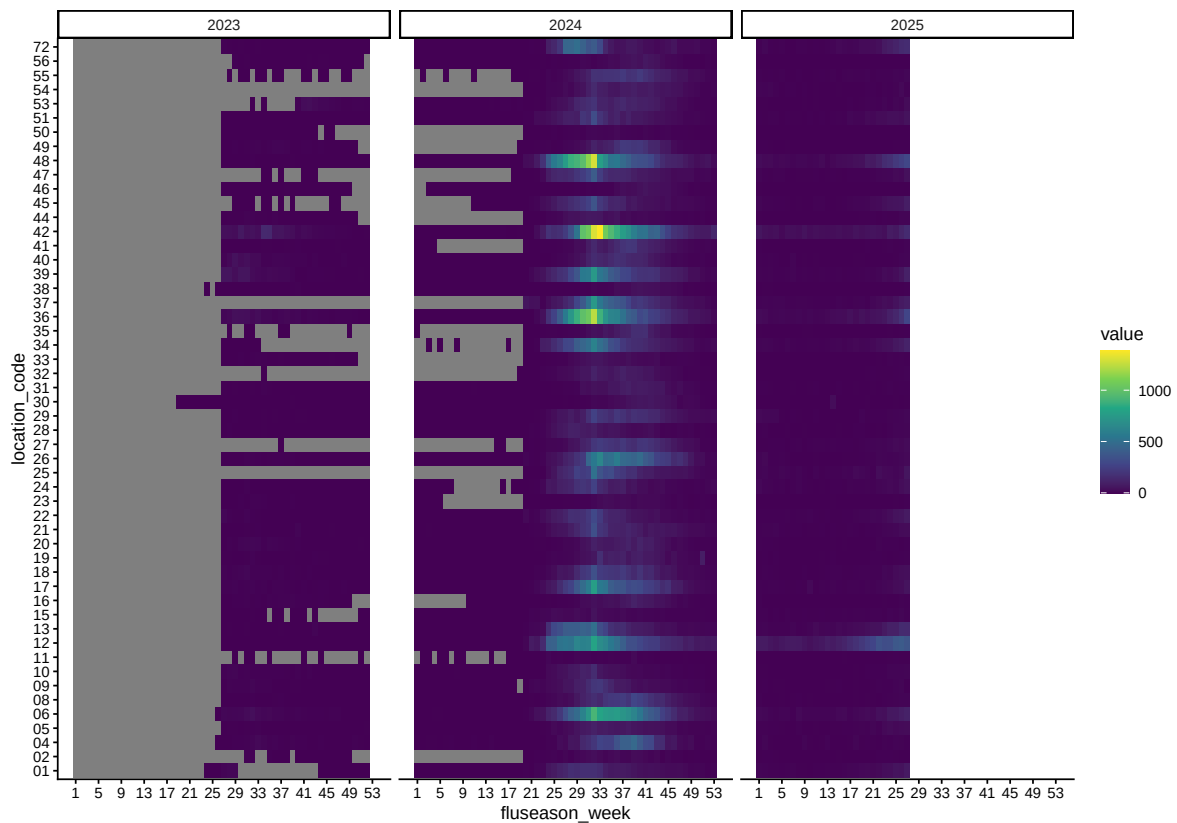
totalconfrsvnewadm: Total number of new hospital admissions of patients with confirmed RSV captured during the reporting week (Sunday - Saturday)

```
library(RSocrata)

nhsn <- read.socrata("https://data.cdc.gov/resource/ua7e-t2fy.json") |>
  select(
    weekendingdate,
    jurisdiction,
    totalconfrsvnewadm
  ) |>
  transmute(
    fluseason = season_year(as.Date(weekendingdate)),
    fluseason_week = season_week(as.Date(weekendingdate)),
    abbreviation = jurisdiction,
    value = as.numeric(totalconfrsvnewadm),
    datasetH1 = "NHSN",
    datasetH2 = "NHSN",
    sample = "1"
  ) |>
  filter(abbreviation != "USA") |>
  left_join(
    location_codes |> select(abbreviation, location_code),
    by = "abbreviation"
  ) |>
  drop_na(location_code) |>
  filter(fluseason >= 2023)

nhsn |>
  ggplot(aes(x = fluseason_week, y = location_code)) +
  facet_wrap(~fluseason) +
  geom_tile(aes(fill = value)) +
  scale_fill_viridis_c()
```

```
scale_x_continuous(breaks = seq(1, 53, by = 4)) +
theme_classic()
```



```
nhsn |>
  skimr::skim() |>
  as_tibble() |>
  select(skim_variable, complete_rate)
```

```
# A tibble: 8 x 2
  skim_variable complete_rate
  <chr>          <dbl>
1 abbreviation      1
2 datasetH1         1
3 datasetH2         1
4 sample           1
5 location_code     1
6 fluseason         1
```



```
7 fluseason_week      1
8 value               0.715
```

```
write_parquet(nhsn, "RSV_files/NHSN.parquet")
```

Modelling Hub

Sample output: Each trajectory must have a unique identifier combining `run_grouping` and `stochastic_run`. Quantile and CDF outputs: There is no stochastic variation. The sample column can also be set to NA or to a consistent identifier that reflects the quantile or epiweek?

```
library(httr2)
library(purrr)

gh_ls <- function(url) {
  request(url) |> req_perform() |> resp_body_json()
}

base <- "https://api.github.com/repos/midas-network/rsv-scenario-modeling-hub/contents/model

parquet_files <-
  gh_ls(base) |>
  keep(~ .x$type == "dir") |>
  map(~ gh_ls(.x$url)) |>
  flatten() |>
  keep(~ grepl("\\.parquet$", .x$name))

data <- parquet_files |>
  set_names(map_chr(
    parquet_files,
    ~ sub(".*/(.*)\\.gz\\.parquet$", "\\1", .x$path)
  )) |>
  map(~ read_parquet(.x$download_url))

library(tidytable)

rsv_smh <- data |>
  imap(
    ~ .x |>
      as_tidytable() |>
      filter(
```

```

    target == "inc hosp",
    age_group == "0-130"
  ) |>
  mutate(
    forecast_date = as.Date(origin_date) + weeks(horizon - 1),
    run_grouping = if ("run_grouping" %in% names(.)) {
      run_grouping
    } else {
      NA_character_
    },
    stochastic_run = if ("stochastic_run" %in% names(.)) {
      stochastic_run
    } else {
      NA_character_
    },
  ) |>
  transmute(
    location_code = location,
    fluseason = season_year(forecast_date),
    fluseason_week = season_week(forecast_date),
    value,
    datasetH1 = "RSV_SMH",
    datasetH2 = paste(.y, scenario_id, sep = "_"),
    sample = case_when(
      output_type == "sample" ~ paste0(run_grouping, "_", stochastic_run),
      output_type == "quantile" ~ paste0("q", output_type_id),
      output_type == "cdf" ~ paste0("p", output_type_id)
    )
  )
) |>
bind_rows() |>
filter(location_code != "US")

# Aggregated plot
plot_data <- rsv_smh |>
  summarise(
    value = mean(value, na.rm = TRUE),
    .by = c(fluseason, datasetH2, location_code, fluseason_week)
  )
plot_data |>
  filter(fluseason == "2024") |>
  ggplot(aes(x = fluseason_week, y = location_code)) +

```

```
facet_wrap(~datasetH2) +
geom_tile(aes(fill = value)) +
scale_fill_viridis_c() +
scale_x_continuous(breaks = seq(1, 53, by = 4)) +
theme_classic()
```



```
write_parquet(rsv_smh, "RSV_files/RSV_SMH.parquet")
```

Converting NSSP to Counts

We converted NSSP data, which records the percentage of emergency department visits due to RSV, into estimated admission counts. Using weeks and locations with NHSN data (which records the counts), we learned a location-specific scaling factor linking the reported percentage to actual counts. The scaling was done using a Poisson generalized linear mixed-effects model (GLMM) with a random intercept for each location. The predicted counts are the observed percentage scaled by the learned location-specific factor.

```

library(lme4)

# Predict counts and merge
nssp <- read_parquet("RSV_files/NSSP.parquet")
nhsn <- read_parquet("RSV_files/NHSN.parquet")

fit_glmm <- inner_join(
  nssp |>
    select(location_code, fluseason, fluseason_week, value),
  nhsn |>
    select(location_code, fluseason, fluseason_week, value),
  by = c("location_code", "fluseason", "fluseason_week")
) |>
  filter(value.y >= 0, value.x > 0) |>
  mutate(log_value = log(value.x)) |>
  (\(x) {
    glmer(
      value.y ~ (1 | location_code) + offset(log_value),
      data = x,
      family = poisson(link = "log")
    )
  })()

nssp_counts <- nssp |>
  filter(value > 0) |>
  mutate(
    log_value = log(value),
    value = predict(fit_glmm, newdata = cur_data(), type = "response")
  ) |>
  select(location_code, fluseason, fluseason_week, value) |>
  right_join(
    nssp |> select(-value),
    by = c("location_code", "fluseason", "fluseason_week")
  )

write_parquet(nssp_counts, "RSV_files/NSSP_COUNTS.parquet")

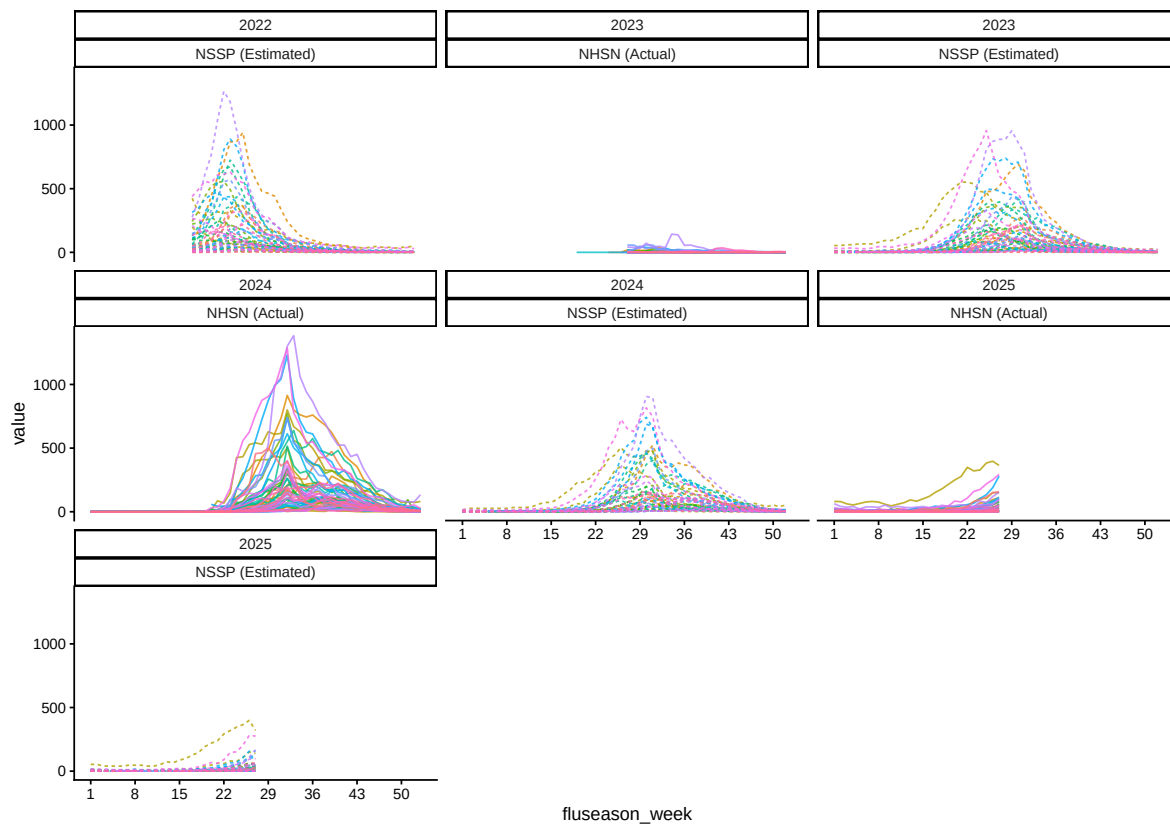
#NHSN vs NSSP Plot
bind_rows(
  nssp_counts |> mutate(source = "NSSP (Estimated)"),
  nhsn |> mutate(source = "NHSN (Actual)")
)

```

```

) |>
  ggplot(aes(
    x = fluseason_week,
    y = value,
    colour = location_code,
    linetype = source
  )) +
  facet_wrap(~ fluseason + source) +
  geom_line(alpha = 0.8) +
  theme_classic()+
  theme(legend.position = "none") +
  scale_x_continuous(breaks = seq(1, 53, by = 7))

```



Summary

Surveillance data includes: - NSSP: Estimated ED visits due to RSV (converted to counts) -> 8350 rows. - RSV-Net: Laboratory-confirmed RSV hospitalizations -> 5502 rows. - NHSN:

Hospital admissions of patients with confirmed RSV -> 6864 rows.

Modeling data includes: - RSV Scenario Modeling Hub: Projected hospitalizations under various scenarios -> 11525100 rows.

```
all_data <- bind_rows(  
  nssp_counts,  
  rsvnet,  
  nhsn,  
  rsv_smh  
)
```